

REPORT DOCUMENT

Lab Task 1:

The image shows a C++ program in a code editor and its execution output in a terminal window. The program defines three variables: an integer 'a' with value 10, a float 'b' with value 3.14, and a char 'c' with value 'Z'. It then prints a table of these variables and their memory addresses. The output shows the variable names, their values, and their addresses in hexadecimal format.

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int a = 10;
5      float b = 3.14;
6      char c = 'Z';
7      cout << "Variable | Value | Address\n";
8      cout << "  a (int) | " << a << " | " << &a << "\n";
9      cout << "  b (float) | " << b << " | " << &b << "\n";
10     cout << "  c (char) | " << c << " | " << (void*)&c << "\n";
11     return 0;
12 }
```

Output:

```
Variable | Value | Address
a (int) | 10 | 0x6ffe3c
b (float) | 3.14 | 0x6ffe38
c (char) | Z | 0x6ffe37
```

Process exited after 0.1176 seconds with return value 0
Press any key to continue . . .

Lab Task 2

top\task2 me.cpp - Dev-C++ 5.11

View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Release

obals)

bug task2 me.cpp

```
1  #include <iostream>
2  using namespace std;
3  void exchangeValues(int *p1, int *p2, int *p3, int *p4) {
4      int temp;
5      temp = *p1;
6      *p1 = *p2;
7      *p2 = temp;
8      temp = *p3;
9      *p3 = *p4;
10     *p4 = temp;
11 }
12 int main() {
13     int a, b, c, d;
14     cout << "Enter 1st integer: "; cin >> a;
15     cout << "Enter 2nd integer: "; cin >> b;
16     cout << "Enter 3rd integer: "; cin >> c;
17     cout << "Enter 4th integer: "; cin >> d;
18     cout << "\nBefore exchange: a=" << a << " b=" << b << " c=" << c << " d=" << d << "\n";
19     exchangeValues(&a, &b, &c, &d);
20     cout << "After exchange: a=" << a << " b=" << b << " c=" << c << " d=" << d << "\n";
21     return 0;
22 }
```

TDM-GCC 4.9.2

(globals)

Classes

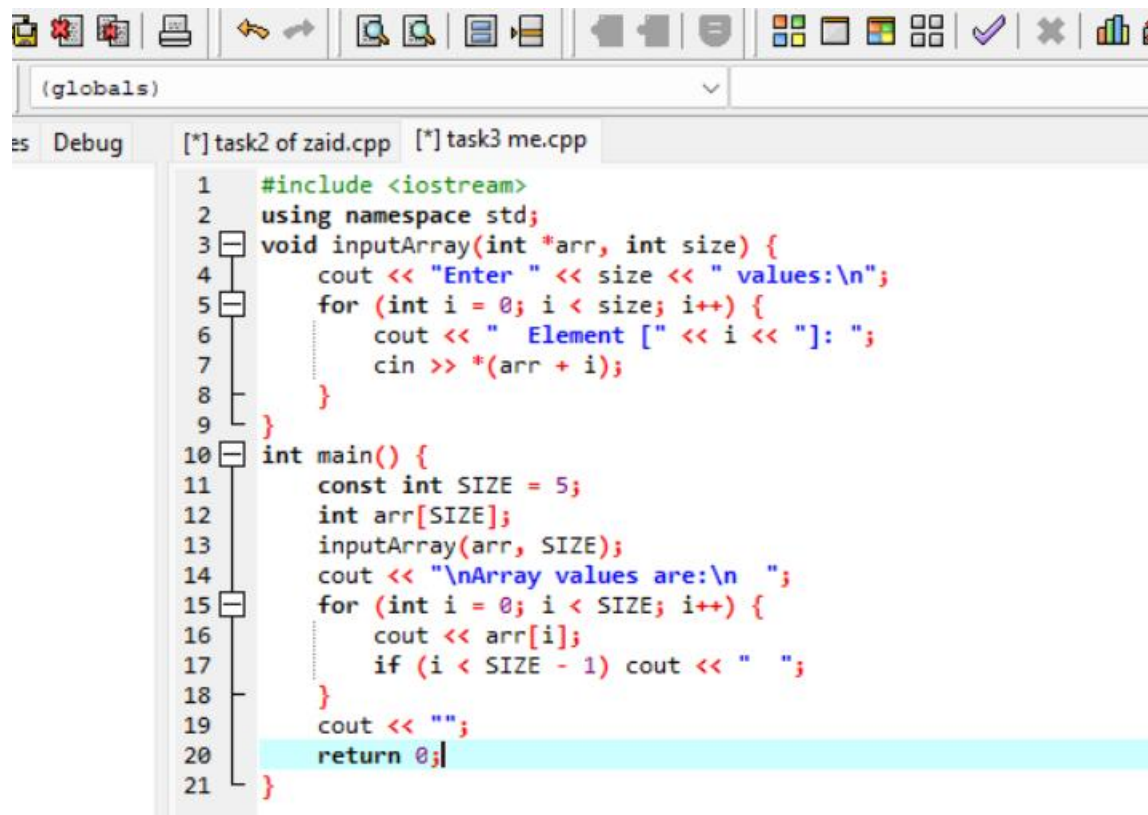
C:\Users\MSI\Desktop\task2 r

```
Enter 1st integer: 5
Enter 2nd integer: 8
Enter 3rd integer: 9
Enter 4th integer: 4
```

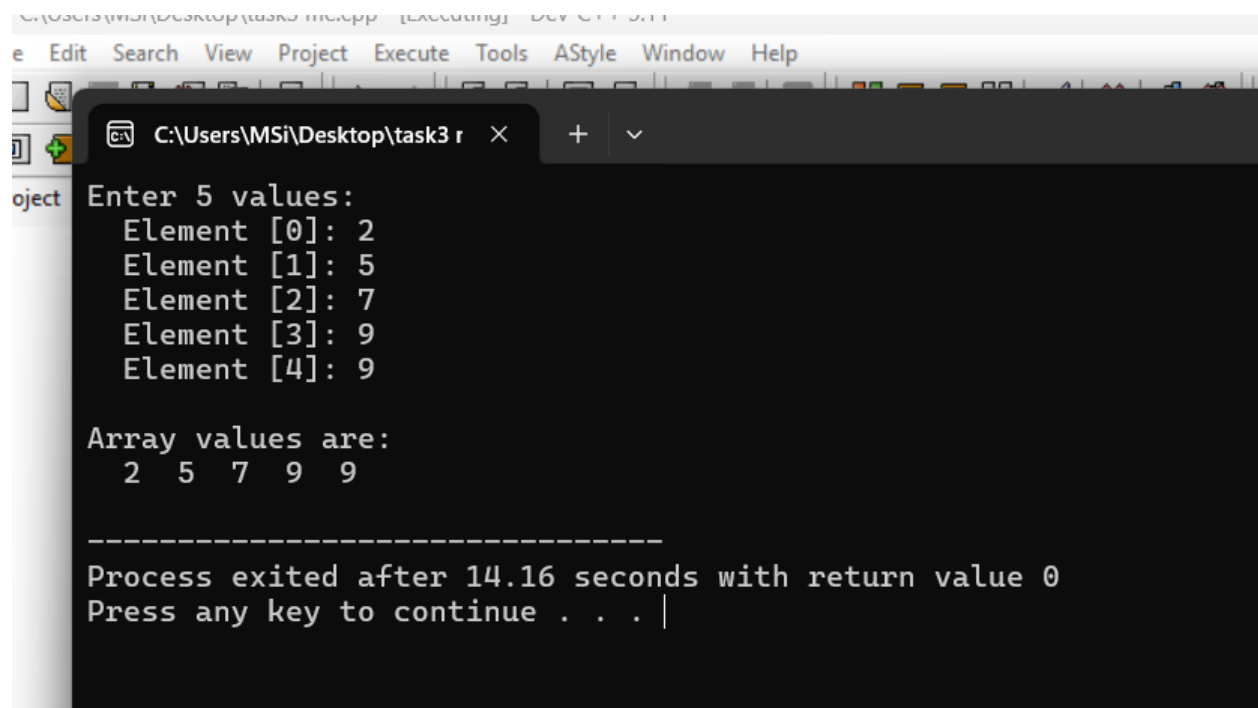
```
Before exchange: a=5 b=8 c=9 d=4
After exchange: a=8 b=5 c=4 d=9
```

```
-----
Process exited after 7.338 seconds with return value 0
Press any key to continue . . .
```

Lab Task 3



```
1  #include <iostream>
2  using namespace std;
3  void inputArray(int *arr, int size) {
4      cout << "Enter " << size << " values:\n";
5      for (int i = 0; i < size; i++) {
6          cout << " Element [" << i << "]: ";
7          cin >> *(arr + i);
8      }
9  }
10 int main() {
11     const int SIZE = 5;
12     int arr[SIZE];
13     inputArray(arr, SIZE);
14     cout << "\nArray values are:\n ";
15     for (int i = 0; i < SIZE; i++) {
16         cout << arr[i];
17         if (i < SIZE - 1) cout << " ";
18     }
19     cout << "\n";
20     return 0;
21 }
```



```
C:\Users\MSi\Desktop\task3 r x + v
e Edit Search View Project Execute Tools AStyle Window Help
object Enter 5 values:
Element [0]: 2
Element [1]: 5
Element [2]: 7
Element [3]: 9
Element [4]: 9

Array values are:
2 5 7 9 9

-----
Process exited after 14.16 seconds with return value 0
Press any key to continue . . . |
```

Lab Task 4

```
1  #include <iostream>
2  using namespace std;
3  void Largest(int *a, int *b, int *c) {
4      int largest = *a;
5      if (*b > largest) largest = *b;
6      if (*c > largest) largest = *c;
7      cout << "Largest = " << largest << "\n";
8  }
9  void Smallest(int *a, int *b, int *c) {
10     int smallest = *a;
11     if (*b < smallest) smallest = *b;
12     if (*c < smallest) smallest = *c;
13     cout << "Smallest = " << smallest << "\n";
14 }
15 void Average(int *a, int *b, int *c) {
16     double avg = (*a + *b + *c) / 3.0;
17     cout << "Average = " << avg << "\n";
18 }
19 int main() {
20     int x, y, z, choice;
21     cout << "Enter three integers:\n";
22     cout << "  Number 1: "; cin >> x;
23     cout << "  Number 2: "; cin >> y;
24     cout << "  Number 3: "; cin >> z;
25     do {
26         cout << "\n----- MENU ----- \n";
27         cout << "1. Find Largest\n";
28         cout << "2. Find Smallest\n";
29         cout << "3. Find Average\n";
30         cout << "4. Exit\n";
31         cout << "Enter choice: ";

```

```
Debug  Untitled1  task4 me new.cpp

C:\Users\MSi\Desktop\task4 r  ×  +  v

Enter three integers:
  Number 1: 3
  Number 2: 5
  Number 3: 7

----- MENU -----
1. Find Largest
2. Find Smallest
3. Find Average
4. Exit
Enter choice: 3
Average = 5
```

• HOME TASKS

Home Task 1

```
home task 1 me.cpp

1  #include<iostream>
2  using namespace std;
3  int maxNum(int* ptr, int size)
4  {
5      int max = *ptr;
6      for(int i = 1; i < size; i++)
7      {
8          if(*(ptr + i) > max)
9              max = *(ptr + i);
10     }
11     return max;
12 }
13 int main()
14 {
15     int arr[] = {34, 7, 89, 12, 56, 43};
16     int size = sizeof(arr) / sizeof(arr[0]);
17     cout << "Maximum: " << maxNum(arr, size) << endl;
18     return 0;
19 }
```

```
C:\Users\MSI\Desktop\home · × + ▾
Maximum: 89
-----
Process exited after 0.1221 seconds with return value 0
Press any key to continue . . . |
```

Home task 2

```
home task 1 me.cpp  home task 2 me.cpp
1  #include<iostream>
2  using namespace std;
3  void arrReverse(int* ptr, int size)
4  {
5      int* start = ptr;
6      int* end = ptr + size - 1;
7      while(start < end)
8      {
9          int temp = *start;
10         *start = *end;
11         *end = temp;
12         start++;
13         end--;
14     }
15 }
16 int main()
17 {
18     int arr[] = {1, 2, 3, 4, 5};
19     int size = 5;
20     arrReverse(arr, size);
21     for(int i = 0; i < size; i++)
22         cout << arr[i] << " ";
23     return 0;
24 }
```

```
C:\Users\MSI\Desktop\home · × + ▾
5 4 3 2 1
-----
Process exited after 0.11 seconds with return value 0
Press any key to continue . . . |
```

Home task 3

```
1  #include <iostream>
2  using namespace std;
3  int countVowels(char* str) {
4      char* ptr = str;
5      int count = 0;
6      int index = 0;
7      cout << "\nVowels found:\n";
8      while (*ptr != '\0') {
9          char ch = *ptr;
10         if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
11             ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U') {
12             cout << " Vowel: " << ch << " at index " << index << "\n";
13             count++;
14         }
15         ptr++;
16         index++;
17     }
18     return count;
19 }
20 int main() {
21     char str[]="Programming";
22     int total = countVowels(str);
23     cout << "\nTotal vowels = " << total << "\n";
24     return 0;
25 }
```

```
C:\Users\MSi\Desktop\extens x + v

Vowels found:
Vowel: 'o' at index 2
Vowel: 'a' at index 5
Vowel: 'i' at index 8

Total vowels = 3

-----
Process exited after 0.1372 seconds with return value 0
Press any key to continue . . . |
```

Home task 4

```
home task 4 me.cpp
1  #include<iostream>
2  using namespace std;
3  void StringCopy(char* dest, char* src)
4  {
5      while(*src != '\0')
6      {
7          *dest = *src;
8          dest++;
9          src++;
10     }
11     *dest = '\0';
12 }
13 int main()
14 {
15     char source[] = "Hello, Pointers!";
16     char destination[50];
17     StringCopy(destination, source);
18     cout << "Copied: " << destination << endl;
19     return 0;
20 }
```



```
C:\Users\MSI\Desktop\home' x + v
Copied: Hello, Pointers!

-----
Process exited after 0.1226 seconds with return value 0
Press any key to continue . . .
```

Home Task 5

```
home task 5 me.cpp
1  #include<iostream>
2  using namespace std;
3  void bubbleSort(int* arr, int n)
4  {
5      for(int i = 0; i < n - 1; i++)
6      {
7          for(int j = 0; j < n - i - 1; j++)
8          {
9              if(*(arr + j) > *(arr + j + 1))
10             {
11                 int temp = *(arr + j);
12                 *(arr + j) = *(arr + j + 1);
13                 *(arr + j + 1) = temp;
14             }
15         }
16     }
17 }
18 int main()
19 {
20     int arr[] = {63, 24, 11, 21, 10};
21     int n = 5;
22     cout << "Before: ";
23     for(int i = 0; i < n; i++)
24         cout << *(arr + i) << " ";
25     bubbleSort(arr, n);
26     cout << "\nAfter: ";
27     for(int i = 0; i < n; i++)
28         cout << *(arr + i) << " ";
29     return 0;
30 }
```

```
C:\Users\MSI\Desktop\home × + v
Before: 63 24 11 21 10
After: 10 11 21 24 63
-----
Process exited after 0.1331 seconds with return value 0
Press any key to continue . . . |
```

Home Task 6

```
[*] task6_palindrome.cpp
1  #include <iostream>
2  using namespace std;
3  int myStrLen(char* s) {
4      char* p = s;
5      while (*p != '\0') p++;
6      return (int)(p - s);
7  }
8  bool isPalindrome(char* s) {
9      int len = myStrLen(s);
10     char* left = s;
11     char* right = s + len - 1;
12     while (left < right) {
13         if (*left != *right) return false;
14         left++;
15         right--;
16     }
17     return true;
18 }
19 char toLow(char c) {
20     if (c >= 'A' && c <= 'Z') return c + 32;
21     return c;
22 }
23 bool isLetter(char c) {
24     return (c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z');
25 }
26 bool isPalindromeExtended(char* s) {
27     int len = myStrLen(s);
28     char* left = s;
29     char* right = s + len - 1;
30
31     while (left < right) {
32         ...
```

```
C:\Users\MSI\Downloads\task × + v
Basic Palindrome Check:
"madam" => Palindrome
"racecar" => Palindrome
"hello" => Not Palindrome
"level" => Palindrome

Extended Check (ignore spaces & case):
"madam" => Palindrome
"racecar" => Palindrome
"hello" => Not Palindrome
"level" => Palindrome
"A man a plan a canal Panama" => Palindrome

Enter your own string to check: tawasal
Result (extended): Not Palindrome

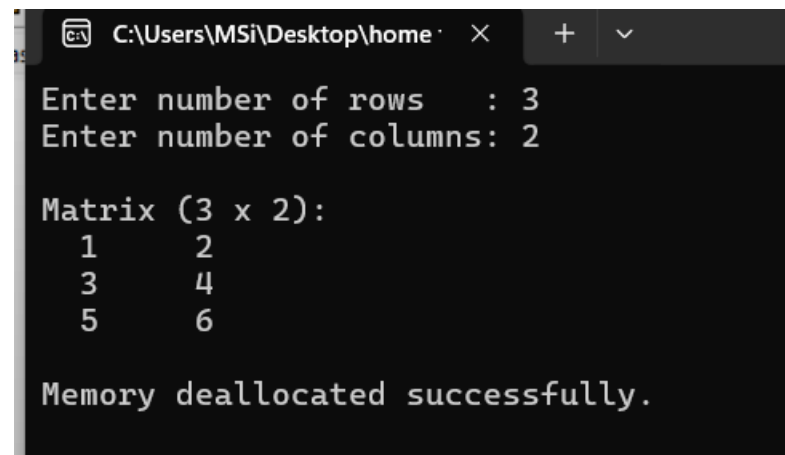
-----
Process exited after 11.63 seconds with return value 0
Press any key to continue . . . |
```

Home Task 7

```

1  #include <iostream>
2  using namespace std;
3  int main() {
4      int rows, cols;
5      cout << "Enter number of rows : "; cin >> rows;
6      cout << "Enter number of columns: "; cin >> cols;
7      int** matrix = new int*[rows];
8      for (int i = 0; i < rows; i++) {
9          matrix[i] = new int[cols];
10     }
11     for (int i = 0; i < rows; i++) {
12         for (int j = 0; j < cols; j++) {
13             matrix[i][j] = i * cols + j + 1;
14         }
15     }
16     cout << "\nMatrix (" << rows << " x " << cols << "):\n";
17     for (int i = 0; i < rows; i++) {
18         cout << " ";
19         for (int j = 0; j < cols; j++) {
20             cout << matrix[i][j];
21             if (j < cols - 1) cout << "\t";
22         }
23         cout << "\n";
24     }
25     for (int i = 0; i < rows; i++) {
26         delete[] matrix[i];
27     }
28     delete[] matrix;
29     cout << "\nMemory deallocated successfully.\n";
30     return 0;
31 }

```



```

C:\Users\MSI\Desktop\home >
Enter number of rows : 3
Enter number of columns: 2

Matrix (3 x 2):
1      2
3      4
5      6

Memory deallocated successfully.

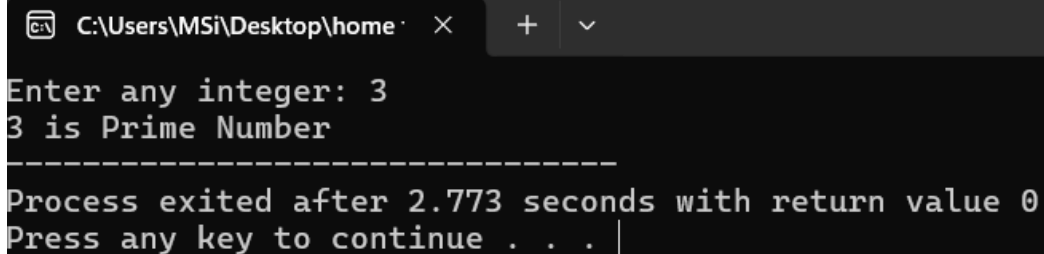
```

Home Task 8

```

1  #include<iostream>
2  using namespace std;
3  void Prime(int *num)
4  {
5      int count = 0;
6      for(int i = 1; i <= *num; i++)
7      {
8          if(*num % i == 0)
9          {
10             count++;
11         }
12     }
13     if(count == 2)
14         cout << *num << " is Prime Number";
15     else
16         cout << *num << " is Not Prime Number";
17 }
18 int main()
19 {
20     int a;
21     cout << "Enter any integer: ";
22     cin >> a;
23     Prime(&a);
24     return 0;
25 }

```



```

C:\Users\MSi\Desktop\home >
Enter any integer: 3
3 is Prime Number
-----
Process exited after 2.773 seconds with return value 0
Press any key to continue . . .

```

Home Task 9

```

2  using namespace std;
3  void inputArray(int *arr, int size)
4  {
5      cout << "Enter "<< size << " values:\n";
6      for(int i = 0; i < size; i++)
7      {
8          cin >> *(arr + i);
9      }
10 }
11 void doubArray(int *arr, int size)
12 {
13     for(int i = 0; i < size; i++)
14     {
15         *(arr + i) = *(arr + i) * 2;
16     }
17 }
18 int main()
19 {
20     int arr[3];
21     inputArray(arr, 3);
22     cout << "Original Array: ";
23     for(int i = 0; i < 3; i++)
24     {
25         cout << *(arr + i) << " ";
26     }
27     doubArray(arr, 3);
28     cout << "\nDoubled Array(new): ";
29     for(int i = 0; i < 3; i++)
30     {
31         cout << *(arr + i) << " ";
32     }

```

```

C:\Users\MSI\Desktop\home
Enter 3 values:
1 2 3
Original Array: 1 2 3
Doubled Array(new): 2 4 6
-----
Process exited after 2.545 seconds with return value 0
Press any key to continue . . .

```